

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Fifth Semester B.E. (CSE) (Credit System) Degree Examinations**

January 2023

20CS502 – DATABASE MANAGEMENT SYSTEMS

Duration: 3 Hours

Max. Marks: 100

*Note: Answer **Five full** questions choosing **One full** question from **each Unit**.***Unit – I**

	Marks	BT*	CO*	PO*
1. a) Briefly explain the various characteristics of the database approach.	10	L*2	1	1
b) Design an ER diagram and schema for the flight management system. Specify any assumptions made.	10	L3	1	3
2. a) With a neat flow diagram explain the main phases involved in the database design process.	10	L2	1	1
b) Explain the following with example for each: i. Structural constraints ii. Recursive relationships	10	L2	1	1

Unit – II

3. a) Consider the Sailors-Boats-Reserves DB described Below SAILOR (sid, sname, rating, age) BOAT (bid, bname, color) RESERVES(sid, bid, date) Write each of the following queries in Relational Algebra. i. Find all sailor id's of sailors who have a rating of at least 8 or reserved boat 103. ii. Find the names of sailors who have reserved a red and a green boat. iii. Find the sids of sailors with age over 20 who have not reserved a red boat..	8	L3	2	1
b) Illustrate the following relational algebra operations with example for each. i. Division ii. Natural join	8	L2	2	1
c) Explain any 4 aggregate functions used in relational algebra. Give example for each.	4	L1	2	1
4. a) Consider the following Relations: Consider the database schema given below. Employee (name, ssn, address, sex, salary, superssn) Project (pname, pnumber, plocation) Works_On (essn, pno, hours) Specify the following queries in relational algebra i. Retrieve the names of all female employees who do not work on any project. ii. Find the names of all employees who are directly supervised by 'Franklin Wong'. iii. For each project on which more than 5 employees work, list the project name and the total hours per week (by all employees) spent on that project.	8	L3	2	2
b) Explain the ER-to-Relational Mapping Algorithm with example.	8	L2	2	1
c) Explain grouping concepts in relational algebra. Give one example.	4	L2	2	2

P.T.O.

Unit – III

5. a) Consider the following database for a banking enterprise:
 BRANCH (branch-name: string, branch-city: string, assets: real)
 ACCOUNT (accno: int, branch-name: string, balance: real)
 DEPOSITOR (customer-name: string, accno: int)
 CUSTOMER (customer-name: string, customer-street: string, customer-city: string)
 LOAN (loan-number: int, branch-name: string, amount: real)
 BORROWER (customer-name: string, loan-number: int)
 1. Find all the customers who have at least 2 accounts at all the branches located in a specific city.
 2. Find all the customers who have accounts in at least 1 branch located in all the cities
 3. Find all the customers who have accounts in at least 2 branches located in a specific city
 12 L2 3 1
 b) How basic constraints are specified in SQL-99. Explain with example. Give syntax.
 8 L3 3 1
6. a) Consider the following database of student enrollment in courses & books adopted for each course:
 STUDENT (regno: string, name: string, major: string, bdate: date)
 COURSE (course #: int, cname: string, dept: string)
 ENROLL (regno: string, course#: int, sem: int, marks: int)
 BOOK_ADOPTION (course#: int, sem: int, book-ISBN: int)
 TEXT (book-ISBN: int, book-title: string, publisher: string, author: string)
 i) Produce a list of text books (include Course #, Book-ISBN, Book-title) in the alphabetical order for courses offered by the 'CS' department that use more than two books.
 ii) List any department that has all its adopted books published by a specific publisher.
 iii) List the book ISBNs and book titles of the department that has maximum number of students.
 12 L3 3 1
 b) What is a view? Give the specification of view in SQL.
 8 L2 3 1

Unit – IV

7. a) Consider the following two sets of FDs.
 $F = \{A \rightarrow C, AC \rightarrow D, E \rightarrow AD, E \rightarrow H\}$
 $G = \{A \rightarrow CD, E \rightarrow AH\}$
 Check whether they are equivalent?
 10 L3 4 3
 b) List and explain the various informal design Guidelines for Relation Schemas.
 10 L2 4 1
8. a) What is the need for normalization? Explain first, second and third normal form by giving example for each.
 10 L2 4 3
 b) Consider the universal relation $R = \{A, B, C, D, E, F, G, H\}$ and the set of Functional Dependencies:
 $F = \{A \rightarrow B, ABCD \rightarrow E, EF \rightarrow G, EF \rightarrow H, ACDF \rightarrow EG\}$
 Find a minimal cover G for F.
 10 L3 4 2

Unit – V

9. a) Explain Strict Two-Phase Locking protocol in detail.
 10 L2 5 1
 b) With an example, Explain aggregate data models of NoSQL.
 10 L2 5 2
10. a) Illustrate the anomalies associated with Interleaved Execution. Explain in detail.
 10 L2 5 2
 b) What is CAP theorem? Explain in detail.
 10 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
